

Unconcentrated markets, persistent ties: incumbency in Türkiye’s public IT procurement

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Abstract

Market-level concentration indices are the standard first screen for competition problems, yet a procurement market can be unconcentrated while its buyers keep returning to the same suppliers. We study this gap with 9,991 awarded information-technology (IT) contracts from Türkiye’s e-procurement platform EKAP (2010–2026), linking 2,814 firms to 2,890 public buyers. After a value-based audit of the IT scope, firm-name resolution, and deflation, count-based Herfindahl–Hirschman indices are below 1,000 in every product market. Buyers, however, return to incumbent suppliers far more often than market structure implies. Of contracts awarded by a buyer that had bought in the same market before, 45% went to an incumbent, against 21% under a permutation null that fixes each buyer’s demand and each firm’s wins within product market, year, and province. About a third of these contracts are annual re-tenders of the same service, and they account for most of the excess, but excess incumbency persists for new needs. In a supplier-choice model with firms active before each tender, prior supply to the buyer raises a firm’s odds of winning about 40-fold. A placebo with future suppliers indicates that persistent buyer–firm matching explains much of this, with the clearest state dependence in renewals. Incumbency is highest in health information systems and under the negotiated procedure without publication, where it rose sharply in 2017–18, and incumbent-won tenders attract single bids far more often. Buyer-level incumbency, benchmarked against null models, complements market concentration in procurement monitoring.

Keywords: Public procurement, Incumbency, Switching costs, Market concentration, Supplier choice, Türkiye

JEL Classification: H57 , L14 , L86 , D85

1 Introduction

Competition authorities and procurement regulators usually begin with market structure. The Herfindahl–Hirschman index (HHI) and concentration ratios summarize how supply is shared among the firms in a relevant market, and fixed thresholds turn them into a first judgement about competitive risk (Hirschman, 1964; Rhoades, 1993; U.S. Department of Justice and Federal Trade Commission, 1992, 2023). In public procurement this screen has a blind spot. A market can contain many suppliers, none with a large share, while individual public buyers keep returning to the same firm. Market-level shares are then low, but the effective choice set of each buyer may be narrow. This is the familiar logic of switching costs and aftermarkets: once a customer has installed a system, competition for its later purchases can be weak even when competition for new customers is strong (Borenstein, MacKie-Mason, & Netz, 2000; Farrell & Klemperer, 2007; Farrell & Shapiro, 1988; Klemperer, 1987; Shapiro, 1995). Public IT is a natural case, and Greenstein (1993, 1995) showed long ago that installed base shapes vendor choice in US federal computer procurement. Persistent ties can, however, also reflect stable fit between a buyer’s needs and a supplier’s products, or relational contracting that rewards reliable suppliers (Board, 2011; Calzolari & Spagnolo, 2009). Distinguishing these explanations is a classic problem in the study of state dependence (Dubé, Hitsch, & Rossi, 2010; Heckman, 1981).

This paper measures the gap between market-level concentration and buyer-level persistence for the public IT market of Türkiye, and asks how much of the persistence each explanation can account for. We reconstruct 9,991 awarded IT contracts from the national e-procurement platform EKAP (2010–2026), linking 2,814 firms to 2,890 buyers. We ask three questions. Is the market concentrated once it is measured carefully? Do buyers return to incumbent suppliers more often than market structure, timing, and geography would produce? And where, for which kinds of purchases, and under which procedures is incumbency strongest?

The answers are as follows. First, the market is not concentrated. Count-based HHIs are below 1,000 in every product market. Value-based HHIs exceed 1,000 only in three small markets, each dominated by one or two very large contracts, and in two of them the index falls below 1,000 once the largest contract is removed. Second, incumbency is pervasive. Of the contracts awarded by a buyer that had already bought in the same product market, 45.2% went to a firm that had supplied that buyer in that market before. Permutation nulls reassign winners within cells while keeping each buyer’s demand and each firm’s number of wins fixed. They predict 6.2% when cells are product market and year, and 20.7% when cells also fix the buyer’s province. Third, a large part of incumbency is re-tendering: about a third of these contracts renew a service bought by the same buyer 6–18 months earlier, with an incumbency rate of 72%. New needs still go to incumbents about twice as often as the province null predicts. In a conditional-logit model of supplier choice among firms active before each tender, having supplied the buyer raises the odds of winning about 40-fold. Firms that will supply the buyer only in the future, however, have almost the same advantage. Much of the persistence therefore reflects stable buyer–firm matching; the state dependence that switching costs imply is clearest in renewals. Fourth, incumbency is highest in health information systems and under Article 21(b) of Law No. 4734, which

allows negotiation without prior publication. There, the gap relative to open tenders rose sharply in 2017–18, when hospital procurement was reorganized. Incumbent-won tenders attract a single bid far more often than tenders won by other firms.

We contribute in three ways. First, recent work has combined concentration measures with network analysis of procurement markets and shown that concentration indices and centrality can point in different directions (I. Fountoukidis, Daffi, Antoniou, & Varsakelis, 2026; I.G. Fountoukidis, Antoniou, & Varsakelis, 2023; Pliatsidis, 2024). We move from the structure of the network to the persistence of its ties, and we benchmark that persistence against explicit null models and a supplier-choice model with a future-supplier placebo. This separates relational persistence from the arithmetic dilution of pooled indices and from the sorting of firms by market, time, and geography, and it shows how much of the persistence is compatible with switching costs. Second, we decompose incumbency into the renewal of recurring services and the award of new needs, a distinction that matters for interpreting incumbency as lock-in and that is rarely drawn in procurement data. Third, we provide the first network-based evidence on Turkish public IT procurement, adding to work that has focused on political connections and construction (Gürakar, 2016; Gürakar & Bircan, 2019; Kutlay & Yıldırım, 2026). We also document a value-based audit of keyword-defined procurement data that is relevant to other users of EKAP.

2 Background

2.1 Switching costs, incumbency, and discretion

The HHI is informative only relative to a relevant market; pooled across markets that are not substitutes, it is mechanically diluted (Rhoades, 1993). In procurement the relevant choice set can be narrower still. A hospital that has installed an information system faces data-migration, training, and integration costs if it changes supplier. The switching-cost and aftermarket literatures show how such costs let a supplier with modest market share hold substantial power over its installed base (Borenstein et al., 2000; Farrell & Shapiro, 1988; Klemperer, 1987; Shapiro, 1995; Shapiro & Varian, 1999), and the European Commission has warned of this lock-in in public IT procurement (European Commission, 2013). Empirically, observed inertia can reflect switching costs (true state dependence) or persistent heterogeneity in preferences and fit, and the two are hard to separate (Dubé et al., 2010; Heckman, 1981; Shcherbakov, 2016).

In procurement auctions, incumbents often hold cost or information advantages that shape entry and bidding (De Silva, Dunne, & Kosmopoulou, 2003; Jofre-Bonet & Pesendorfer, 2003; Li & Zheng, 2009), and limited competition in government purchasing is common (Kang & Miller, 2022). Buyer discretion can amplify or discipline incumbency. More discretion raises the share of repeat winners without necessarily worsening performance (Coviello, Guglielmo, & Spagnolo, 2018). It can also be used to favour connected firms (Baltrunaite, Giorgiantonio, Mocetti, & Orlando, 2021; Palguta & Pertold, 2017; Szucs, 2024), and it can be sustained by relationships that develop with a buyer’s tenure (Coviello & Gagliarducci, 2017). Relational contracts can make loyalty to proven suppliers valuable (Board, 2011; Calzolari & Spagnolo,

2009). Past performance legitimately informs supplier choice, although it may deter entry (Butler, Carbone, Conzo, & Spagnolo, 2020; Decarolis, Spagnolo, & Pacini, 2016), and negotiation can be efficient for complex purchases (Bajari, McMillan, & Tadelis, 2009). Repeated contracting is therefore not by itself evidence of harm; its welfare consequences depend on prices and quality (Bandiera, Prat, & Valletti, 2009). The empirical questions we can answer are how much incumbency exists beyond what market structure implies, for which purchases, and how much of it is compatible with state dependence.

2.2 Procurement as a network

Contract-award records are naturally two-mode. Network studies of procurement have used this structure to study corruption risk and favouritism (Fazekas & Wachs, 2020; Wachs, Fazekas, & Kertész, 2021), to detect collusion in co-bidding networks (Reeves-Latour & Morselli, 2017; Wachs & Kertész, 2019), and to describe buyer-supplier transactions (Popa, 2019). The integrity literature builds risk indicators from award data, most prominently single bidding and the concentration of a buyer's spending on one supplier (Fazekas & Kocsis, 2020; Fazekas, Tóth, Abdou, & Al-Shaibani, 2024; Fazekas, Tóth, & King, 2016). Closest to our question, I.G. Fountoukidis et al. (2023) show for the European pacemaker market that concentration-type indices can signal competition while centrality reveals dominant suppliers, and Pliatsidis (2024) measures concentration in Greek procurement with network methods by product group. We share their premise that shares and network structure can diverge. We differ in the object of analysis, the buyer-supplier tie rather than node centrality, and in benchmarking it against nulls and a choice model.

2.3 The Turkish context

Turkish public procurement is governed by Law No. 4734 (2002) and administered by the Public Procurement Authority (KİK), whose platform EKAP publishes notices and awards (Alsaç, 2017); see Buğra and Savaşkan (2014) and Gürakar (2016) for its political economy. Article 21 lists the grounds for the negotiated procedure. Article 21(b) permits it where a tender must be held urgently because of unforeseen events such as natural disasters or epidemics. In these cases, and under Article 21(f), the buyer need not publish a notice and instead invites at least three firms of its choosing. Law No. 7144 (in force 25 May 2018) added to 21(b) cases with special features of construction technique or urgency for structural safety, which concern works rather than IT (Alyanak, 2022). Article 21(f) covers purchases of goods and services below an annually indexed monetary limit. Politically connected firms have been found to win more often under less competitive procedures (Gürakar, 2016; Gürakar & Bircan, 2019), and market capture intensified during the 2010s (Kutlay & Yıldırım, 2026). These studies focus mainly on construction and on political connections. The health sector was reorganized during our window: Decree-Law (KHK) No. 694 of 25 August 2017 abolished the Public Hospitals Authority and its provincial hospital unions and returned hospital procurement to the Ministry of Health and to individual hospitals.

3 Data

3.1 Collection and scope

We collected award records of IT-related tenders from EKAP by keyword search and retained those concluded with an identifiable winner (13,418 records). EKAP search matches substrings of titles, so it admits tenders in which an IT term appears only incidentally. The keyword *ERP*, for example, matches place names and *tuz serpici* (salt spreader), and *technical support* matches hospital staffing contracts. An earlier count-based cleaning removed 394 obvious false positives, leaving 13,024 contracts, but a value-based audit showed that the remaining errors were concentrated among large contracts. We therefore classified the 150 largest contracts by hand, applied ordered title rules to the rest, and inspected random samples from every rule. Of the 13,024 contracts, 1,627 (12.5%, holding 17.7% of nominal value) are not IT, including housing construction, mining overburden removal, vehicles, and cleaning services, and a further 1,406 are ambiguous, such as hospital data-entry staffing or building works that include security cameras. The main sample contains the 9,981 unambiguous IT contracts and 10 contracts for IT-enabled call-centre services. One of these, the 2025 call-centre contract for the national hospital appointment system (MHRS), is by far the largest in the data; all value-based results are reported with and without it. The ambiguous contracts form a robustness sample. EKAP’s award list reports one winner per tender, so multi-lot tenders appear with a single winner; about 116 titles suggest multiple lots, and we found no framework-agreement call-offs.

3.2 Entities, markets, and values

Firms appear on EKAP under spelling and legal-form variants. We canonicalized names by normalizing characters and stripping legal-form suffixes, and reviewed the merge groups of the 300 largest firms by hand, including firms that changed legal form or name under the same brand. Joint ventures were kept separate. This reduces 4,788 recorded names to 4,389 firms, 2,814 of them in the main sample. Some buyers are recorded under generic labels that pool units across provinces, most notably the provincial health directorates of the Ministry of Health. We split such labels by the province recorded for each tender (34 labels covering 732 main-sample contracts) and use the recorded labels as a sensitivity check. We assign each contract to one of 14 product markets from its title and each buyer to one of 11 sectors from its name. A blind re-classification of 300 random titles, produced with AI assistance and not yet checked by human coders, agreed with the rule-based market in 77% of cases (Cohen’s $\kappa = 0.74$), with precision and recall of 0.99 for health information systems. Values are deflated to 2025 lira with the annual consumer price index; results with the producer price index are in the Supplementary Information. Of the 2,814 firms, 302 are identifiable as natural persons (sole proprietors); in the released data, all winners that are natural persons or lack an explicit legal-entity form are pseudonymized. Tables 1 and 2 describe the main sample.

Table 1 Main sample (IT and IT-enabled services, EKAP 2010–2026).

Awarded contracts	9,991
Firms (canonicalized)	2,814
Buyers (pooled labels split by province)	2,890
Buyer–firm pairs	7,219
Pairs with two or more contracts	1,377 (19.1%)
Contracts in repeated pairs	41.5%
Total value, 2025 TRY (nominal TRY)	89.3 bn (33.3 bn)
Largest contract (MHRS call centre, 2025), share of real value	8.4%
Procedure: open / Art. 21(b) / Art. 21(f) / other	59.0 / 10.5 / 29.6 / 0.9%
Tender dates	Oct 2010 – Mar 2026

Table 2 Product markets (main sample). Value in bn 2025 TRY; procedure shares in % of contracts.

Product market	Contracts	Firms	Buyers	Value	21(b)	21(f)
Health information systems	2,189	205	935	19.48	34.7	9.4
ERP / management software	1,515	599	770	11.14	4.5	47.1
Custom software / web / mobile	1,117	734	625	7.04	2.9	43.2
Software licences	1,047	375	445	7.68	1.8	23.4
Network / data-centre infrastructure	748	390	420	7.43	5.9	25.0
Computers / peripherals	696	467	415	2.85	3.9	27.7
Maintenance / support services	626	297	251	8.23	3.5	33.2
GIS / city information	553	203	320	4.75	2.2	41.4
Cybersecurity	390	194	229	3.25	5.4	33.1
Physical security / surveillance	385	222	203	2.29	1.8	23.1
Other IT	277	215	165	1.82	5.4	27.8
Education technology	205	128	124	1.87	5.4	50.2
Smart city / traffic	169	106	111	3.44	4.1	34.3
Call centre / help desk	74	46	55	7.99	5.4	51.4
All	9,991	2,814	2,890	89.26	10.5	29.6

4 Methods

4.1 Concentration

For each product market we compute the supplier HHI, $H = \sum_i s_i^2$ on a 0–10,000 scale, and the four-firm ratio, from contract counts and from real values. Because single contracts can dominate value shares, we also report the value HHI without the market’s largest contract, the largest change from removing any single contract, and a version with values winsorized at the 99th percentile. Confidence intervals come from 1,000 bootstrap resamples of contracts within market; because the plug-in HHI is biased upward under resampling, we use bias-shifted percentile intervals. We also compute HHIs within calendar years. As reference points we use 1,000 and 1,800, the boundaries of the “moderately” and “highly” concentrated ranges in the 1982 and 1992 US merger guidelines, the second of which the 2023 guidelines reinstated ([U.S.](#)

Department of Justice and Federal Trade Commission, 1992, 2023). These are merger-screening conventions, not measures of conduct. Finally, we decompose the pooled index as $H_{\text{agg}} = \sum_m w_m^2 H_m + C$, where w_m is market m 's share and C collects the terms of firms active in several markets, to show how much of the gap between pooled and market-level concentration is arithmetic.

4.2 Incumbency, renewals, and null models

Let contract i be awarded by buyer k in product market m on date t . Contract i is *repeat-eligible* if k awarded at least one other contract in m strictly before t , and *incumbent* if its winner had won a contract from k in m strictly before t . The incumbency rate is the share of repeat-eligible contracts that are incumbent. A repeat-eligible contract is a *renewal* if the same buyer awarded a contract in the same market 6–18 months earlier whose title is similar; otherwise it is a *new need*. Titles count as similar if the Jaccard similarity of their normalized, stemmed words is at least 0.5, after years, numbers, and generic procurement terms are removed. In an AI-assisted check of 42 pairs above the threshold, 38 described the same recurring service. Pairs just below it were mixed, so some renewals are classified as new needs, which likely biases the contrast between the two groups towards zero.

To separate incumbency from market structure, we randomly permute winners across contracts within cells and recompute all statistics. This preserves each buyer's demand and each firm's number of wins in every cell. Firm entry and exit, specialization, and market size are therefore held fixed, and only the assignment of winners to buyers is randomized. Buyer histories, and hence eligibility and renewal status, are unaffected. In null N1 the cells are product market $\times$ calendar year. N2 adds the buyer's sector, and N3 instead adds the buyer's province. N4 adds the winner's home province, defined as the province where it won most contracts before the tender (firms without earlier wins form their own category), and N4b adds both provinces. Each null uses 1,000 permutations with fixed seeds. With 1,000 draws the smallest attainable one-sided p -value is 0.001, which we report as $p < 0.001$ when no permutation reaches the observed value. Subgroups are compared with the same permutations restricted to the subgroup. For buyer-level dependence we compare, for each of the 540 buyers with at least five contracts, the main supplier's share and the supplier HHI with the buyer's own null distribution, and control the false discovery rate across buyers (Benjamini & Hochberg, 1995).

4.3 Supplier choice and state dependence

The nulls randomize winners among all firms active in a cell. A sharper benchmark compares the winner with plausible alternatives that existed before the tender. For each repeat-eligible contract we form a choice set of the winner and the firms that won at least one contract in the same market in the 24 months before the tender date. We then estimate a conditional logit (McFadden, 1974); random subsets of alternatives, which we use as robustness checks, preserve consistency (McFadden, 1978). Contracts won by firms without a win in the market in the previous 24 months (entrants) cannot be represented in this choice set and are analysed separately. The alternative-specific covariates are:

- whether the firm previously supplied the buyer in the same market (incumbent);
- whether it supplied the buyer in another market;
- whether its home province, computed from prior wins, is the buyer’s province;
- the logs of its wins in the market over the previous 12 months, of its total prior wins, and of its prior wins with buyers of the same sector.

Standard errors are clustered by buyer.

A large incumbent coefficient is compatible both with state dependence and with persistent match heterogeneity, where some firms simply fit some buyers better (Dubé et al., 2010; Heckman, 1981). We use two diagnostics. First, a placebo adds an indicator for firms that will supply the buyer in the market only after the tender date. Match heterogeneity predicts a large coefficient for such future suppliers as well; pure state dependence predicts a much smaller one. Because the mechanics of the design inflate the future-supplier coefficient even without heterogeneity, we benchmark it with 100 simulations in which winners are generated by a model with state dependence only. Second, we separate the buyer’s most recent supplier in the market from earlier suppliers, since switching costs should favour the most recent one.

4.4 Procedure and timing

We model the probability that a repeat-eligible contract is won by an incumbent with a logit. Covariates are the procedure (open and restricted procedures as reference), an indicator for tenders from 25 May 2018 and its interaction with 21(b), the log real value, and the buyer’s prior contracts, prior suppliers, and years of history in the market. The model includes product-market, year, and buyer-sector fixed effects, and standard errors are clustered by buyer and winning firm (Cameron, Gelbach, & Miller, 2011). Because the timing of any change is uncertain, we also estimate an event study of the 21(b)–open gap by year, for all buyers and for health buyers with and without buyer fixed effects, using 2013–17 as the reference period.

4.5 Single bidding

Bid counts are not part of EKAP’s award lists. We retrieved them from result announcements for a year-stratified random sample of tenders. In 393 main-sample tenders from 2012–2026 the number of bids is reported; we weight by the inverse sampling fraction of each year and report design-based intervals. A second, larger but non-random source covers 1,194 tenders from 2021–2026 whose titles contain the word *yazılım* (software) and reports the number of valid bids.

5 Results

5.1 Markets are not concentrated

Figure 1 shows supplier concentration by product market. Pooled over all markets, the supplier HHI is 39 by count and 149 by real value (92 without the MHRS contract). By count, every product market is below 1,000; the highest values are for call-centre services (690; 74 contracts) and health information systems (431, 95% CI 405–458).

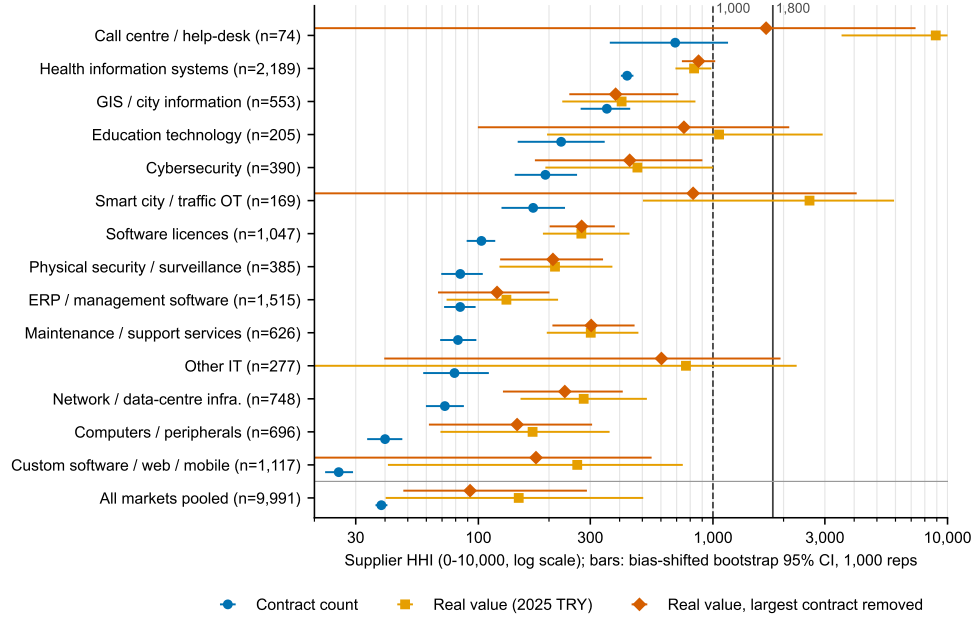


Fig. 1 Supplier concentration by product market: HHI by contract count, by real value, and by real value without the market’s largest contract, with bias-shifted bootstrap 95% intervals (1,000 resamples). Lines mark 1,000 and 1,800. Log scale.

By real value, three markets exceed 1,000. Removing the largest contract reduces the value HHI of smart-city and traffic systems from 2,580 to 823 and of education technology from 1,062 to 752. The call-centre market stays above 1,000 on every value variant, but it consists largely of the MHRS contract. Health information systems, with 2,189 contracts and 205 firms, is the one large market with stable concentration close to the moderate range: its value HHI is 831 (692–985), its value CR4 53%, and removing any single contract moves the index by less than 40 points. Within single years, where small samples inflate all indices, its median count HHI is 671 (597 after bias correction) and its median value HHI is 1,091.

The decomposition shows why pooled and market-level measures differ. By count, $\sum_m w_m^2 H_m$ accounts for 27 of the 39 points of the pooled index, and the weighted mean of market HHIs is 179. Dilution across markets is arithmetic, so we treat market definition as a precondition rather than a result. Even with product markets defined, supply is at most moderately concentrated.

5.2 Buyers return to incumbents

Of the 9,991 contracts, 4,913 were awarded by a buyer that had bought in the same product market before, and 45.2% of these went to an incumbent. Table 3 compares this rate with the nulls. When winners are reassigned within product market and year (N1), the expected rate is 6.2%. Conditioning also on the buyer’s sector (N2) raises it to 11.0%, on the buyer’s province (N3) to 20.7%, and on the winner’s home

Table 3 Incumbency rate against permutation nulls (1,000 permutations each; all $p < 0.001$). Renewals: a contract with a similar title bought by the same buyer in the same market 6–18 months earlier.

	n	Observed	Null mean: winners permuted within market $\times$ year and			
			– (N1)	sector (N2)	buyer prov. (N3)	both prov. (N4b)
All repeat-eligible	4,913	0.452	0.062	0.110	0.207	0.264
Renewals	1,743	0.720	0.092	0.166	0.288	0.391
New needs	3,170	0.305	0.045	0.078	0.163	0.195
no similar earlier title	2,117	0.188	0.034	–	0.104	–
Article 21(b)	618	0.723	0.143	0.159	0.376	–
Open	3,041	0.441	0.057	0.112	0.203	–

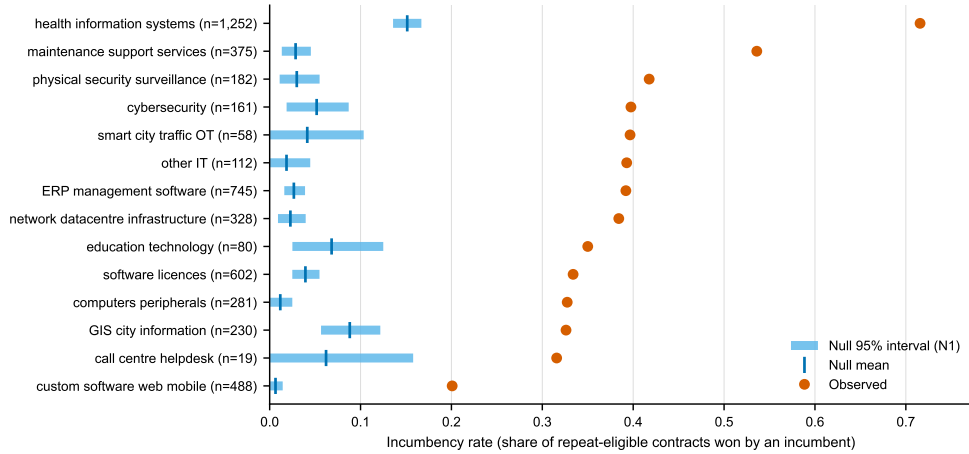


Fig. 2 Incumbency rate by product market: observed (dots) and permutation null N1 (mean and 95% interval; winners permuted within product market $\times$ year). n is the number of repeat-eligible contracts.

province (N4) to 17.3%. Conditioning on both provinces (N4b) raises it to 26.4%, although under N4b 39% of eligible contracts sit in cells with a single possible winner. The observed rate exceeds every null ($p < 0.001$), by about 1,900 contracts under N1 and 920 under N4b. The excess holds in every year from 2011 to 2026 and in every product market (Fig. 2). It is also robust to recorded instead of split buyer labels and to recorded instead of canonicalized firm names. Dropping call-centre tenders, multi-lot tenders, or natural-person winners does not change it, and neither does adding the ambiguous-scope contracts (Supplementary Information).

5.3 Renewals and new needs

A large part of incumbency is the re-tendering of recurring services. The median gap between a buyer's successive contracts in a market is about a year, and 35.5% of repeat-eligible contracts are renewals. Their incumbency rate is 72.0%, against 28.8%

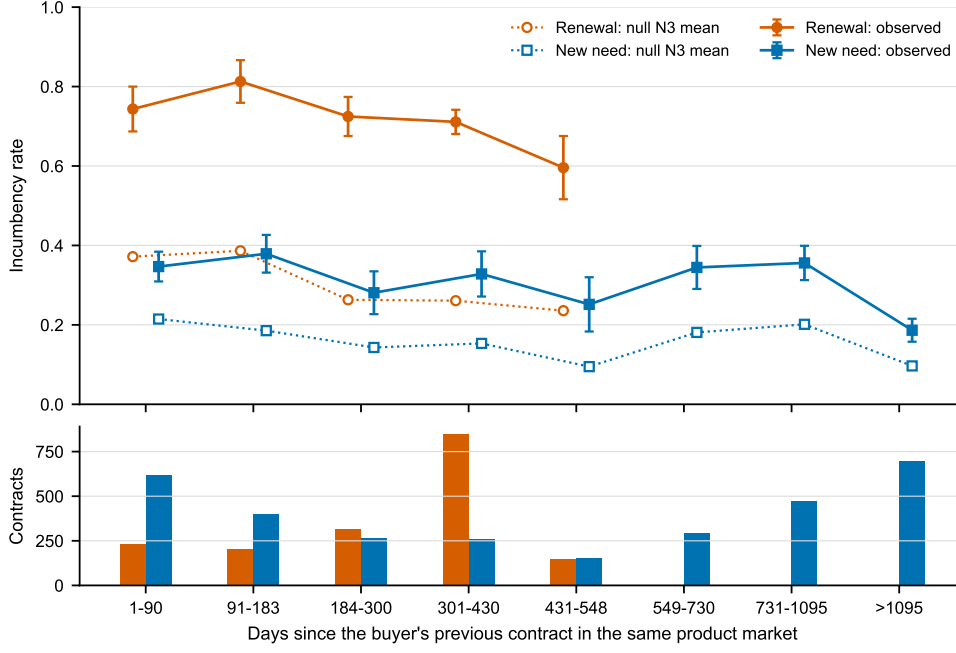


Fig. 3 Incumbency by days since the buyer's previous contract in the same product market, for renewals and new needs: observed (95% intervals) and null N3 mean (winners permuted within product market $\times$ year $\times$ buyer province). Lower panel: number of contracts.

under N3, and they account for 57–64% of the excess, depending on the null (Table 3). Incumbency is not only renewal, however. New needs go to incumbents in 30.5% of cases, against 16.3% under N3 and 19.5% under N4b. Even contracts with no similar earlier title at all go to incumbents 1.8 times as often as N3 predicts. Figure 3 shows that the difference between renewals and new needs holds at every interval since the buyer's previous contract, and that new needs stay above their null throughout. The pattern holds in each of the eight largest markets. In health information systems, 87% of renewals and 61% of new needs go to incumbents (N3 nulls 45% and 38%); in enterprise software the figures are 82% and 23% (N3 nulls 23% and 7%). About one in ten repeat-eligible contracts renews a licence or maintenance agreement for a named product, where only the vendor or its dealers may be able to supply. Excluding these contracts leaves incumbency essentially unchanged (renewals 2.4 and new needs 1.9 times their N3 null).

5.4 Supplier choice and state dependence

Table 4 reports the conditional logit. In 61% of repeat-eligible contracts the winner had won in the market in the previous 24 months and so belongs to the choice set; in the rest the winner is an entrant, which is itself evidence that incumbency is far from absolute. Among the contracts with established winners, having supplied the buyer in

Table 4 Conditional logit of supplier choice among firms active in the market in the 24 months before each tender. Odds ratios with buyer-clustered 95% confidence intervals; 3,013 contracts with established winners (909 buyers).

	Odds ratio [95% CI]
Main model	
Incumbent: supplied this buyer in this market before	42.3 [36.4, 49.2]
Supplied this buyer in another market	2.17 [1.68, 2.81]
Home province (from prior wins) = buyer's province	2.85 [2.44, 3.33]
Log wins in this market, previous 12 months	1.89 [1.71, 2.10]
Log total prior wins	0.91 [0.83, 1.01]
Log prior wins with buyers of the same sector	1.79 [1.60, 2.00]
Incumbent odds ratio by subgroup	
Renewals / new needs	140.7 / 21.0
Article 21(b) / open	95.8 / 33.7
All contracts, entrant winners added (lower bound)	21.5 [18.5, 25.0]
Placebo and recency	
Past supplier / future supplier only	63.7 / 54.4 [42.2, 70.1]
Future supplier only, simulated pure state dependence	19.8 [16.1, 24.5]
Most recent supplier / earlier supplier	45.8 / 26.2
Renewals: past / future supplier only	258 / 128
New needs: past / future supplier only	30 / 40

the same market raises a firm's odds of winning 42-fold (buyer-clustered 95% CI 36–49). This holds constant the firm's recent activity in the market, its overall experience, its experience with buyers of the same sector, and whether it is based in the buyer's province. The estimate is stable across choice-set sizes (10 or 30 random alternatives: 44 and 42) and a 12-month activity window (40). It is largest for renewals (141) and under Article 21(b) (96), but it is also large for new needs (21) and open tenders (34). When entrant winners are added back to the choice sets, the odds ratio is 22 (19–25). This is a lower bound, because entrants cannot be distinguished from the sampled alternatives.

The placebo qualifies the interpretation. Firms that will supply the buyer in the market only after the tender date also have a large advantage (odds ratio 54, 95% CI 42–70), almost as large as that of past suppliers (64 in the same model). In simulations with state dependence only, the design yields a future-supplier odds ratio of 20 (95% interval 16–25). The observed value is well above this, which points to persistent buyer–firm matching: some firms fit some buyers' needs, locations, or systems better, and would be chosen repeatedly even without switching costs. State dependence is nonetheless present. Past suppliers are favoured far more than future suppliers in renewals (258 against 128). The buyer's most recent supplier is favoured over earlier suppliers, with odds ratios of 46 and 26 and a ratio of 1.75 (1.43–2.14), a recency pattern consistent with switching costs. For new needs, past and future suppliers have similar advantages (30 and 40). The persistence of ties in new purchases thus looks more like matching than lock-in.

5.5 Dependent buyers

The excess is not driven by a few large buyers. Among the 540 buyers with at least five contracts, the median share of the main supplier is 0.33, against a typical null value of 0.16 (Supplementary Fig. S2). After controlling the false discovery rate at 5%, 51% of these buyers have a main-supplier share and 66% a supplier HHI above their own null (39% and 54% under N2). A quarter award at least half of their IT contracts to one firm. Dependence is most pronounced among individual Ministry of Health hospitals and dental centres, 88% of which exceed their null. Per contract, excess incumbency is somewhat lower among the largest buyers (0.34 against 0.45 in the smallest tercile), although their tighter null distributions make a larger share of them individually distinguishable from the null.

5.6 Procedure and timing

Incumbency is higher under Article 21(b) than under open tendering (72.3% against 44.1%), and the gap of 28 points exceeds its null gap under N1–N3 ($p < 0.001$). Most excess incumbency nonetheless occurs in open tenders (61% of the excess under N1), because they are far more numerous. Among 21(b) contracts, 90% of renewals go to the incumbent. Under Article 21(f), the other route without a published notice, 34.6% of repeat-eligible contracts go to an incumbent, 2.6 times the N3 null.

The 21(b)–open gap changed over time (Fig. 4). Relative to 2013–17, when it did not differ significantly from zero (about -7 points for all buyers), it jumped in 2018. The jump was 37 percentage points for all buyers (95% CI 24–50) and 43 points for health buyers (23–62). Part of the jump persists: from 2019 the gap averages 14 points above its 2013–17 level for all buyers ($p = 0.003$) and 12 points for health buyers ($p = 0.04$). With buyer fixed effects, the persistent increase among health buyers is 24 points (10–38). Three caveats apply.

- The timing is imprecise. The best-fitting break is August 2017 rather than May 2018, but only 18 repeat-eligible 21(b) contracts fall between the two dates. A placebo break in May 2016 is also significant in the contract-level logit (odds ratio 2.10). In a specification whose pre-period includes 2010–12, a period with little buyer history and a high gap, the within-buyer change is smaller and imprecise (7 points, $p = 0.20$).
- The change is concentrated among health buyers. The $21(b) \times \text{post-2018}$ interaction has an odds ratio of 2.44 ($p = 0.003$) for health buyers and 0.89 for others. The difference between them is not statistically significant (triple interaction 2.74, 95% CI 0.79–9.51), because only 84 non-health 21(b) contracts are repeat-eligible.
- Under 21(b) the buyer chooses which firms to invite, so the procedure itself makes an incumbent’s participation likely.

The 2018 amendment concerns works, and the change is concentrated in health, where individual hospitals took over procurement in 2017. We therefore read the pattern as a descriptive shift in how, and by which buyers, the exception route was used after the hospital reorganization, not as the causal effect of a legal change.

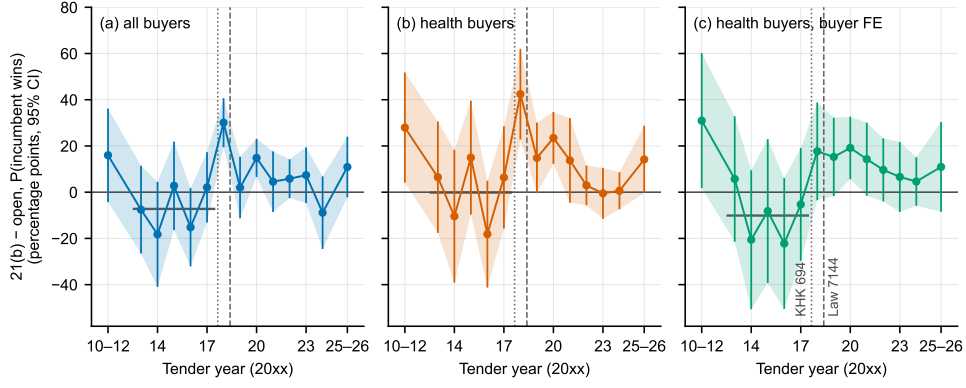


Fig. 4 Difference in the probability that an incumbent wins between Article 21(b) and open tenders, by year. Linear probability models with product-market, year, and buyer-sector fixed effects (buyer fixed effects in panel c), log value, and buyer-history controls; two-way clustered 95% intervals. Horizontal bars: 2013–17 reference mean. Dotted and dashed lines: KHK 694 (25 August 2017) and Law 7144 (25 May 2018).

5.7 Incumbency and single bidding

Incumbent wins coincide with weak competition at the tender stage. In the random sample, 44% of tenders drew a single bid (95% CI 39–49%): 75% under 21(b) and 41% under open procedures (Fig. 5). Tenders won by an incumbent drew a single bid in 71% of cases (61–80%), against 32% (24–42%) for tenders won by a firm that had not supplied the buyer in that market. Adjusting for procedure, year, and renewal status, the odds ratio of a single bid is 3.6 for incumbent winners, and 3.0 for a single valid bid; the gap is present in open tenders as well. In the larger 2021–26 software sample, the odds ratio of a single valid bid for incumbent winners is 2.7 with the same controls. Renewals won by newcomers are the most contested tenders (29% single valid bids). In tenders where the number of firms that obtained the documents is recorded, on average four firms did so and fewer than two submitted a valid bid. These are associations. Incumbents may win because others do not bid, or others may not bid because the incumbent is expected to win.

5.8 Network structure

The network itself is sparse and only moderately modular. Communities in its giant component exceed a degree-preserving bipartite null by a small margin (unweighted modularity 0.73 against 0.69). They align partly with buyer sectors and weakly with product markets, and brokerage across communities is largely a matter of firm size (Supplementary Information). The relational feature that stands out is not segmentation but the persistence of individual buyer–supplier ties.

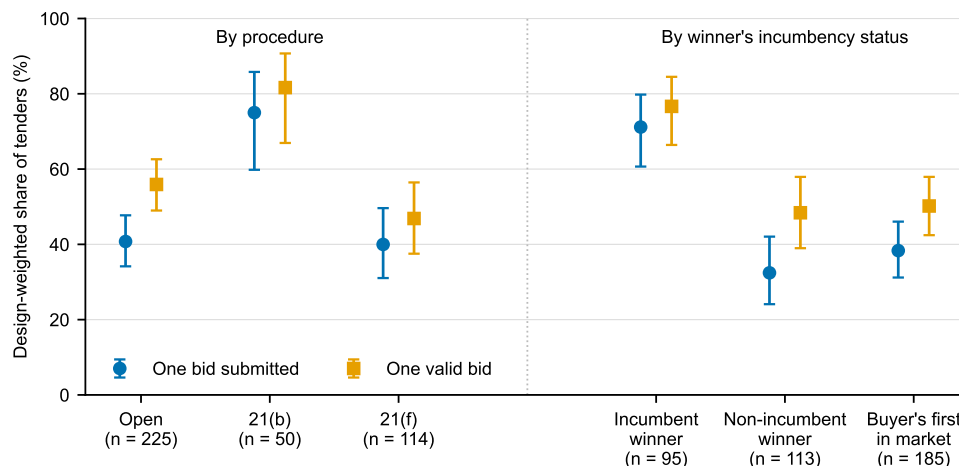


Fig. 5 Single-bid rates by procedure and by whether the winner was an incumbent (design-weighted, year-stratified sample of 393 tenders, 2012–2026; 95% intervals).

6 Discussion

6.1 Interpretation

The two levels of analysis give different answers. Across the whole market, many firms compete, no product market is concentrated on robust measures, and pooled indices are very low. Within buyers, relationships persist far beyond what market structure, timing, and geography imply. The findings are consistent. Market shares describe how supply is spread across all buyers, whereas incumbency describes how narrow each buyer's choice is in practice. A market in which each buyer is tied to a different supplier has low concentration and high persistence at once. This is the configuration the aftermarket literature describes, although, as we show, persistence need not mean lock-in (Borenstein et al., 2000; Shapiro, 1995).

The evidence speaks to the competing explanations in different ways. Switching costs from installed systems predict three patterns. Incumbency should be concentrated in renewals and system-specific products, the most recent supplier should be favoured over earlier ones, and past suppliers should be favoured more than future ones. All three hold for renewals, which carry most of the excess. For new needs, however, future suppliers are as favoured as past ones. That pattern fits persistent matching between buyers and suppliers, such as specialization, geography, or compatibility with systems in place, better than lock-in. Relational contracting and past performance (Board, 2011; Decarolis et al., 2016) predict persistence of the same kind, and they are also compatible with our findings. An erosion of competition predicts fewer bids when incumbents win, and incumbent-won tenders do attract single bids far more often. This association is consistent with incumbency advantages that

discourage entry, but also with purchases that only the incumbent can supply. We cannot tell whether other firms were deterred or uninterested, and distinguishing these mechanisms would require prices or estimated costs, which our data do not contain.

Procedure is one channel, not the source. Article 21(b), formally an urgency ground, has the highest incumbency, and 90% of its renewals go to the incumbent. Its gap to open tendering widened after the 2017 hospital reorganization, when individual hospitals took over procurement. Yet most excess incumbency occurs in open tenders, and new needs show excess incumbency under every procedure.

6.2 Implications for monitoring

Procurement monitoring relies on market-level statistics and on red flags such as single bidding and a buyer’s spending share with one supplier (Fazekas & Kocsis, 2020; Fazekas et al., 2024, 2016). Indicators of the last kind already exist. What our analysis adds is a benchmark and a decomposition. Buyer-level incumbency can be compared with permutation nulls that account for market size, timing, specialization, and geography. It can be split into renewals, where switching costs are most plausible, and new needs, where persistence mostly reflects matching. Such benchmarked indicators identify candidates for review, above all renewals of hospital information systems under Article 21(b). Whether persistence is harmful cannot be read from these indicators alone. Where switching costs are the concern, measures that lower them may do more for competition than additional tendering. Examples are data-portability and interoperability requirements for health information systems, or separate procurement of licences and maintenance (European Commission, 2013).

6.3 Limitations

The analysis is descriptive. We observe winners, bid counts for samples, and award values, but not losing bidders, estimated costs, or quality. We therefore cannot measure the price or welfare consequences of incumbency, and our separation of state dependence from matching rests on a placebo and a simulation benchmark rather than on a structural model. The bid-count samples are small or non-random and start in 2012. Scope, product markets, and renewals are classified from titles. The classifications were validated on samples, but those validation labels were produced with AI assistance and have not yet been checked by independent human coders. Firm names were canonicalized without tax identifiers, and splitting pooled buyer labels by province is an approximation. Reverting to recorded firm names lowers observed incumbency (0.394, 8.4 times the N1 null), and reverting to recorded buyer labels raises it (0.462, 4.4 times); the excess persists in both cases. Multi-lot tenders appear with a single winner. Finally, the timing evidence cannot separate the 2017 health reorganization from the 2018 legal amendment or from other changes around that time.

7 Conclusion

Türkiye’s public IT market is unconcentrated by conventional measures, yet its buyers keep returning to the same suppliers. Among contracts that could go to a returning

supplier, 45% do, about twice the rate implied by the strictest null and far above the rate implied by market structure alone. Much of this is the renewal of recurring services, where the evidence is consistent with switching costs. For new needs, persistence looks more like stable matching between buyers and suppliers. Persistence is strongest in health IT and under the exception route, and it coincides with single bidding. Market-level concentration is therefore an incomplete guide to competition in public procurement. Buyer-level incumbency, benchmarked against appropriate nulls and separated into renewals and new needs, shows where persistence arises and where lock-in is a plausible concern.

Supplementary information. The Supplementary Information contains the scope audit, entity resolution and classification checks, full concentration tables, all null-model results and sensitivities, the renewal classification, the choice models and placebo, the procedure and timing models with the statutory texts of Article 21, the single-bid samples, and the network analysis.

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Declarations

- **Funding.** This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.
- **Competing interests.** The authors have no competing interests to declare.
- **Ethics approval.** Not applicable. The study uses public administrative records; names of natural persons are pseudonymized in the released data.
- **Data and code availability.** Code, derived data, and a data dictionary are available at <https://github.com/XXXX/TenderNet> and archived on Zenodo (<https://doi.org/10.5281/zenodo.XXXXXXX>). The underlying records are published by the Public Procurement Authority on EKAP (<https://ekapv2.kik.gov.tr>).
- **Use of AI tools.** Claude (Anthropic) was used to assist with analysis code, figure preparation, validation coding of samples of tender titles (disclosed in the Data, Methods, and Limitations sections), and language editing. The authors designed the study, checked the code and results, and take full responsibility for the content.
- **Author contributions.** F.A.: conceptualization, methodology, data curation, software, formal analysis, visualization, writing (original draft, review and editing). A.Ş.: data curation, investigation, validation, visualization, writing (review and editing).

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